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A PALAEOMAGNETIC STUDY OF SVECOFENNIAN BASIC ROCKS: MIDDLE PROTEROZOIC CONFIGURATION OF THE FENNOSCANDIAN, LAURENTIAN AND SIBERIAN SHIELDS

J.D.A. PIPER

Department of Geophysics, University of Liverpool, P.O. Box 147, Liverpool L69 3BX (Great Britain)

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The Svecofennian mobile zone occupies the bulk of the Fennoscandian shield and represents terrain subjected to profound tectonic activity and granite intrusion at ~1800 My. This study covers the palaeomagnetism of basic rocks within this belt in Sweden between 65.5 and 67.5°N (gabbros and diorites of Kallax, Niemisel, Sangis, Stora Lulevatten and the Gällivare, Jokkmokk and Voullerim regions) magnetised during uplift and cooling of this belt at ~1750–1550 My. AF and thermal demagnetisation define a consistent sequence of high to low blocking-temperature components identifying a migration of the geomagnetic field during part of this interval. Together with the Rådmansö gabbro–diorite of central-east Sweden (palaeomagnetic pole 201°E, 36°N) these components yield a comparable sequence of palaeopoles to those derived from uplift magnetisations of the contemporaneous Svecokarelian terrain of Finland. The post-tectonic, Uppsala metabasite suite possesses a magnetite-held ("A") remanence comparable to Svecofennian uplift magnetisations from elsewhere; within the aureole of the Almunge alkaline complex this has been largely displaced by a low blocking temperature ("B") remanence, possibly related to a late stage in the Svecofennian uplift cycle. The Hälleforsnäs giant dyke possesses a magnetite-held remanence attributed to initial cooling at 1518 My (palaeomagnetic pole 167°E, 27°N) and at least two high blocking-temperature components. One of these is correlated with the ~1000–800 My Sveconorwegian mobile activity of southwest Sweden; this latter component is represented as the univectorial remanence in dolerite dykes of this age, and sporadically as a secondary component in the adjacent Svecofennian terrain.

The results are compiled with other palaeomagnetic poles from the Fennoscandian shield to derive a generalised apparent polar wandering path for the interval ~1750–1550 My. They define segments of a large loop which agrees closely with uplift magnetisations from the contemporaneous Hudsonian mobile terrain of the Laurentian shield on a single reconstruction derived from Upper Proterozoic (1450–1200 My) palaeomagnetic data. The two shields thus appear to have formed an integral continental unit during the interval 1750–1200 My. A geological reconstruction of the Siberian and Laurentian shields is also tested and found to yield general agreement with the palaeomagnetic evidence. The major geological implications of the collective reconstruction are an alignment of major tectonic trends and a gradual restriction of anorthosite–Rapakivi magmatism between the termination of the ~1800 and ~1100 My mobile episodes.

1. Introduction

The Fennoscandian shield is subdivided into three structural/age provinces (Kratz et al., 1968), which are illustrated in Fig. 1A. The oldest Saamo–Karelian zone includes ancient Archaean nucleii but was locally rejuvenated in Lower Proterozoic times (to ~1900 My); this was the first part of the shield to be uplifted, with deposition of post-tectonic sediments taking place at ~1800 My (Gerling and Polkanov, 1958). The Svecofennian zone (also referred to as Svecokarelian in its northeastern sector) was entirely reworked by mobile activity at ~1800 My, and was uplifted and cooled over periods of up to several hundred My after this. The Sveconorwegian zone of southwest Scandinavia was similarly reworked at

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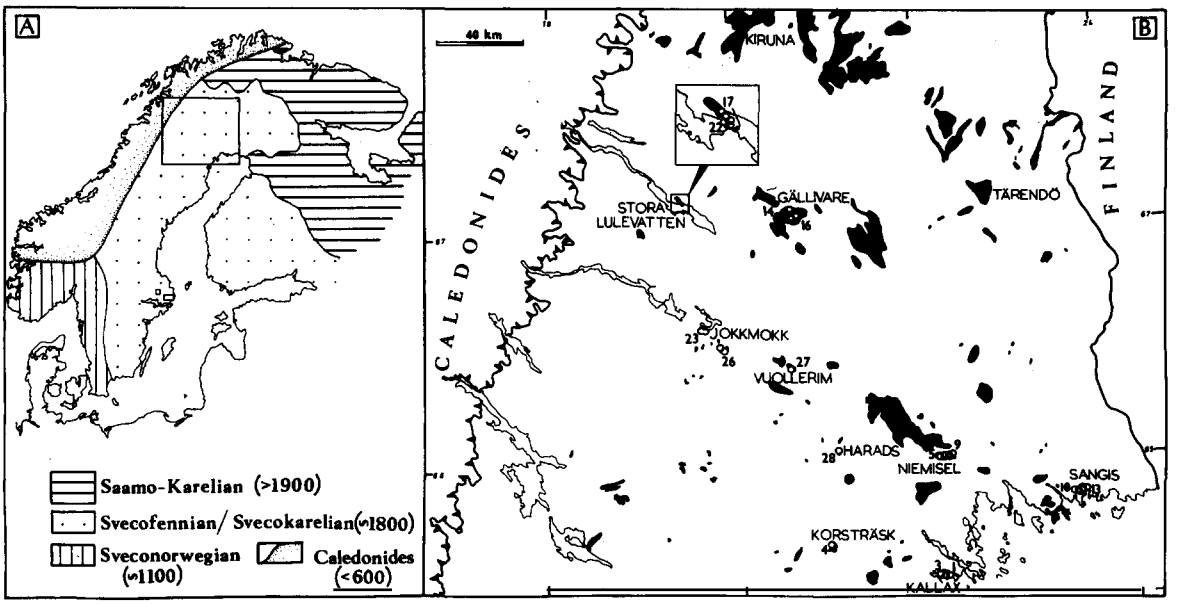


Fig. 1. A: The major tectonic subdivisions of the Fennoscandian shield; the area of the present study in northern Sweden is indicated. B: Palaeomagnetic sampling sites for this study in basic bodies (black) of northern Sweden; the named localities are mentioned in the text.

~1100 My and uplifted and cooled between 1100 and 800 My. Each of these zones is potentially able to yield a record of apparent polar wander (APW) movements relative to the Fennoscandian shield during these intervals of stabilisation and uplift.

The present study covers rocks of the middle (Svecofennian) zone in Sweden and includes rocks stabilised and uplifted between 1750 and 1500 My. The collective data define APW motion during parts of this interval and permit correlation of the Fennoscandian and Laurentian shields in Middle–Upper Proterozoic times.

2. Geology and age

The Svecofennian metamorphic terrain occupies the bulk of the Swedish sector of the Fennoscandian shield (Fig. 1A) and defines a region of infra- and supra-crustal rocks (probably ~2700 and ~2100–1900 My in age respectively) affected by profound mobile activity and granitic intrusion at ~1900–1800 My. Termination of this tectonic activity is defined by the Rb–Sr isochron of 1748 ±40 My* on the late- to post-tectonic Revsund Granite at 64.5°N, 19°E (Welin et al. 1971). In the region covered by the present survey the Svecofennian rocks are predominantly granitic and are grouped together as the "Lina Granite". This is migmatitic granite showing all gradations to country rock (Ödman, 1957) and Rb–Sr studies define ages of 1782 ±50 and 1753 ±60 My (Heier and Compston, 1969; Welin et al., 1971), broadly consistent with U–Pb age estimates of 1845–1775 My from pegmatites cutting the granite (Welin and Blomqvist, 1966). These ages may define the peak of the Svecofennian mobile episode, but it was also followed by a long period of anorogenic magmatism, uplift and cooling. The magmatic activity includes volcanism dated at 1690–1605 My in the Duobblen and Kiruna regions, and syenite and granite intrusions dated between 1690 and 1530 My (Welin et al., 1966; Gulson, 1972). Granitic rocks from near Gällivare (in the vicinity of palaeomagnetic sites 14–

* Ages are adjusted to constants listed by Steiger and Jäger (1977).

16) with an Rb–Sr age of 1530 ±35 My define a magmatic episode called the "Younger Lina Granite" (Welin et al., 1971); the Sjaunjatuottar Granite associated with the Stora–Lulevatten Gabbro and 20 km northeast of palaeomagnetic sites 17–22, yields an Rb–Sr isochron age of 1532 ±25 My (Gulson, 1972). This granitic magmatism may define the end of the Svecofennian cycle in this region.

Palaeomagnetic sampling for this study was concentrated on the gabbros, diorites and amphibolites within a broad zone running westwards from the Baltic coast between 65.5 and 67.5°N (Fig. 1B). They have not been directly dated but are linked with the older and younger Lina Granite episodes, and magnetisations will have a maximum age dating from the uplift and cooling of this mobile terrain. The gabbros at Kallax, Korsträsk, Niemisel and Stora Lullevatten are massive late- or post-tectonic bodies. The basic rocks from Sangis (sites 9–13), Gällivare (14–16), the Jokkmokk area (23–27), and Harads (28) all exhibit some tectonic effects; linear and planar fabrics are locally developed, and the bodies at Sangis, Jokkmokk and Harrads are cut by granite and pegmatite veins. These latter basic rocks may have formation ages comparable to, or older than, the Older Lina Granite, while a significantly younger age may apply to palaeomagnetic sites 1–8 and 17–22.

The history of uplift of the Svecofennian terrain in north Sweden is not yet well-defined, but it was certainly long. Magnusson (1960) reports K–Ar ages of 1770–1700 My from granite rocks of this region but both older and younger ages have been recorded. There are insufficient mineral age data available at present to closely estimate the likely ages of magnetisation in this part of the Svecofennian mobile belt; the best that can be said is that they will be within the interval ~1700–1530 My.

The Rådmansö complex in central-east Sweden, 70 km northeast of Stockholm, is a gabbro–diorite intrusion comparable to many of the bodies sampled in northern Sweden. It has been described in detail by Du Rietz (1929) and intrudes granite and granite–gneiss terrain of Svecofennian age. The gabbro–diorite has not been directly dated but collective radiometric data on associated granites in southeast Sweden yield ages of ~1750 My (Welin, 1966), and K–Ar determinations on granites yield ages in the range 1760–1475 My (Magnusson, 1960). This body has been sampled for palaeomagnetic study at 11 sites.

Svecofennian rocks in the Uppsala region of central Sweden are predominantly granitic but include a suite of post-tectonic metabasite dykes up to several metres in thickness. Although post-dating the Svecofennian tectonism, they were certainly intruded before the end of the thermal–magmatic episodes because they are locally cut by late Svecofennian aplites and pegmatites (Lundegardh, 1957). In addition, all primary mafic minerals have been altered to hornblende, plagioclase is sericitised, and the groundmass is an aggregate of ores, apatite, chlorite and secondary pyrite (Lundegardh, 1957). Fifteen of these dykes were sampled (sites 42–56, Fig. 4) in addition to a related dyke (site 39) intruding the Rådmansö Gabbro–Diorite; nine of these dykes (42–50) lie close to the margin of the later Almunge alkaline complex (1570 My; Kresten et al., 1977).

Later dolerite dykes with an east–west trend crop out sporadically over the region up to 120 km west of Stockholm and include two giant dykes and a variety of smaller dykes. At this western margin they border on rocks affected by the later Sveconorwegian (1100–800 My) thermal magmatic episodes (sites 58 and 59). Samples collected across the thickness of the Hälleforsnäs giant dyke (16.7°E, 59.1°N) comprise site 57. Intrusion of this dyke has been dated by the Rb–Sr method (Patchett, 1978) at 1518 ±38 My. Patchett utilised rheomorphic margin material to date the dyke and noted that the dolerite had experienced extensive pyropylitic alteration either at the time of emplacement or at a later time; feldspar separates suggested a thermal event at ~900 My. Although the Rb–Sr isochrons define an age of ~1500 My for this and associated minor dykes, it is possible that the Svecofennian terrain was still at an elevated temperature at this time. Thus, Patchett (1978) found isotopic evidence at Breven for events between 1485 and 1392 My, and in the Vastevik area, 200 km to the south, Priem and Bakker (1973) determined Rb–Sr and K–Ar mineral ages of 1465–1460 My. It is not clear, however, whether this represents the final stages of Svecofennian uplift and cooling in this area or a local reheating at this later time.

TABLE I

Palaeomagnetic results for Svecofennian basic rocks, northern Sweden

Site	N	D	I	R	k	Palaeomagnetic pole		dp/dm	Curie point(s) (±10°C)	Curve type *	J_0 ($\times 10^{-5}$ A m^2 kg^{-1})
						°N	°E				
(i) Kallax Gabbro (22.0°E, 65.5°N)											
1	6	353	30	5.99	477	40	211	2/3	580, 660	A	0.33–0.02
2(i)	8	317	40	7.95	128	39	257	4/6	575	A	6.21–3.66
(ii)	5	293	35	4.94	72	27	279	6/10			
3	6	347	52	5.98	353	57	222	2/5	575, 630	A	0.77–0.07
(ii) Korstrask Gabbro (20.9°E, 65.7°N)											
4	10	5	47	9.81	48	53	194	6/9	580	A	0.86–0.03
(iii) Niemisel Gabbro (22.1°E, 66.0°N)											
5	7	340	40	6.97	213	45	228	3/5	545	A	6.12–0.41
6	7	336	34	6.96	165	40	231	3/5	550	A	15.03–0.06
7	5	334	36	4.96	103	41	234	5/9	550	A	11.80–2.65
8	5	351	29	4.99	1287	39	212	1/2	560	A	2.29–0.24
Mean, Niemisel Gabbro											
	4	340	35	3.97	110	41	227	6/10			
(iv) Sangis Gabbro (23.5°E, 65.8°N)											
9	7	349	41	6.91	63	47	218	6/9	545	A	0.06–0.02
10	7	9	7	6.97	238	28	192	2/4	540	A	0.02–0.01
11	6	333	–2	5.67	15	8	230	10/19	320, 540	C	0.01–0.00
12	7	350	25	6.97	204	37	216	3/5	560	A	0.13–0.07
13	9	357	28	8.17	10	39	207	10/19	340, 560	C	0.06–0.01
Mean (sites 9, 10, 12 and 13)											
		357	26	3.88	25	38	207	11/20			
(v) Gallivare, basic rocks (20.6°E, 67.1°N)											
14	6	178	–3	5.98	219	–22	23	2/5	560	A	2.51–0.08
15(i)	9	155	41	8.63	21	5	43	8/14	560	A	3.26–0.89
(ii)	5	20	27	4.34	6	34	177	20/37			
16	11	181	50	10.3	15	10	20	11/16	550	A	12.47–0.98
Mean (15(i) and 16)											
(vi) Stora Lulevatten Gabbro (19.4°E, 67.2°N)											
17	7	353	28	6.69	232	38	208	2/4	550	A	4.49–3.11
18(i)	6	355	7	5.89	47	26	203	5/10	560	C	257.92–3.63
(ii)	5	331	33	4.85	26	37	236	10/17			
19	7	0	1	6.74	23	23	200	7/13			28.46–8.32
20	7	0	–5	6.92	73	20	200	4/7	550	A	4.84–2.59
21	6	9	4	5.94	85	24	190	4/7	555	A	9.21–0.14
22(i)	8	352	–3	7.72	25	43	208	6/13			6.69–0.36
(ii)	6	344	37	5.83	30	21	220	8/13			
Mean (18(i), 19, 20, 21, 22(i))											
	5	359	1	4.96	105	23	200	4/8			
Mean (17, 18(ii), 22(ii))											
	3	343	33	2.97	58	40	221	11/19			
(vii) Jokkmokk, basic intrusions (19.6°E, 66.6°N and 19.8°E, 66.5°N)											
23(i)	5	307	–22	4.97	150	3	251	4/6	330, 570	A	0.05–0.01
(ii)	4	308	9	3.95	57	19	256	6/12			
(iii)	5	18	–10	4.86	28	17	181	8/15			
24	9	17	–8	8.76	33	18	182	7/9	–	A	0.01–0.00

TABLE I (continued)

Site	N	D	I	R	k	Palaeomagnetic pole		dp/dm	Curie point(s) (±10°C)	Curve type *	J_0 (× 10^{-5} A m^2 kg^{-1})
						°N	°E				
25(i)	4	4	−24	3.99	684	11	196	2/4	320, 570	A	0.02–0.01
(ii)	5	350	34	4.95	80	41	213	6/10			10.17–3.81
26(i)	4	3	−7	3.91	34	20	197	8/16	330, 570	A	0.65–0.08
(ii)	6	348	32	5.97	189	40	215	3/6			4.31–2.09
Mean (23(iii), 24, 25(i), 26)											
	4	11	−12	3.94	52	17	189	7/13			
Mean (25(ii), 26(ii))											
	2	349	33	1.99	3519	41	214	3/5			
(viii) Vuollerim Gabbro (20.4°E, 66.4°N)											
27(i)	3	359	42	2.95	38	48	202	15/25	555	A	1.77–0.01
(ii)	6	338	4	5.88	40	24	224	5/10			
(ix) Harads, amphibolites (21°E, 66.1°N)											
28	7	3	31	6.00	6	41	197	17/30	–	A	0.06–0.00

* Thermomagnetic curves are defined as: A, heating and cooling curves similar; B, J_s decrease on cooling; C, J_s increase on cooling.

3. Palaeomagnetic results

3.1. Field sampling and laboratory analysis

The palaeomagnetic samples were drilled over several metres of outcrop at each sampling site using a portable drilling motor; they were oriented by sights on the Sun and/or by magnetic compass in conjunction with topographic sights. In the laboratory, samples were measured with an automated parastatic magnetometer and progressively demagnetised in alternating fields in steps of 5 or 10 milliTesla (mT). This treatment was continued to a range of peak fields up to 150 mT. Progressive thermal demagnetisation was undertaken on separate slices of the palaeomagnetic cores or on residual moments in the AF-treated samples in steps of 100 to 500°C and then in steps of 30 to 650°C; this thermal demagnetisation was followed up on the remaining samples where significant directional changes were found. To determine Curie points precisely, thermomagnetic determinations (saturation magnetisation, J_s, versus temperature) were also made on small samples from each site; some representative determinations are illustrated in Fig. 6 and thermomagnetic characteristics are summarised in Tables I and II.

The complete palaeomagnetic observations are listed in Table I. In general, any moment with a coercivity less than 15 mT is considered most likely to be a secondary moment of viscous origin and is not discussed further. As is typical with slowly-cooled basement rocks however, samples frequently exhibit a progressive sequence of directional changes with treatment in higher fields, and directions isolated in this way are listed in order of increasing coercivity or blocking temperature in Table I. A discussion of specific demagnetisation changes follows.

3.2. Svecofennian basic rocks, northern Sweden

Sites 1 to 3 are from a 20 km^2 gabbro exposed at Kallax (22.0°E, 65.5°N) on the Baltic coast southwest of Lulea (Fig. 1). Following initial movement away from the present field (0–15 mT), sites 1 and 3 show no further change in direction or moment to further AF and thermal demagnetisation; since the end-points are stable to 620°C, with 10–15% of initial remanence remaining, it seems probable that the remanence resides in part in a haemo-ilmenite. Site 2 with higher remanent intensities yields a mean (>15 mT) with contrasting declination (D = 316°, I = 40°). With high field treatment, 5 samples migrate further to D = 293°, I = 35° and exhibit an additional fall in moment (Fig. 2) and direction. 2(i) in Table I may be a composite of this end point and sites 1 and 3. Samples from the 12 km^2 gabbro at Korstrask

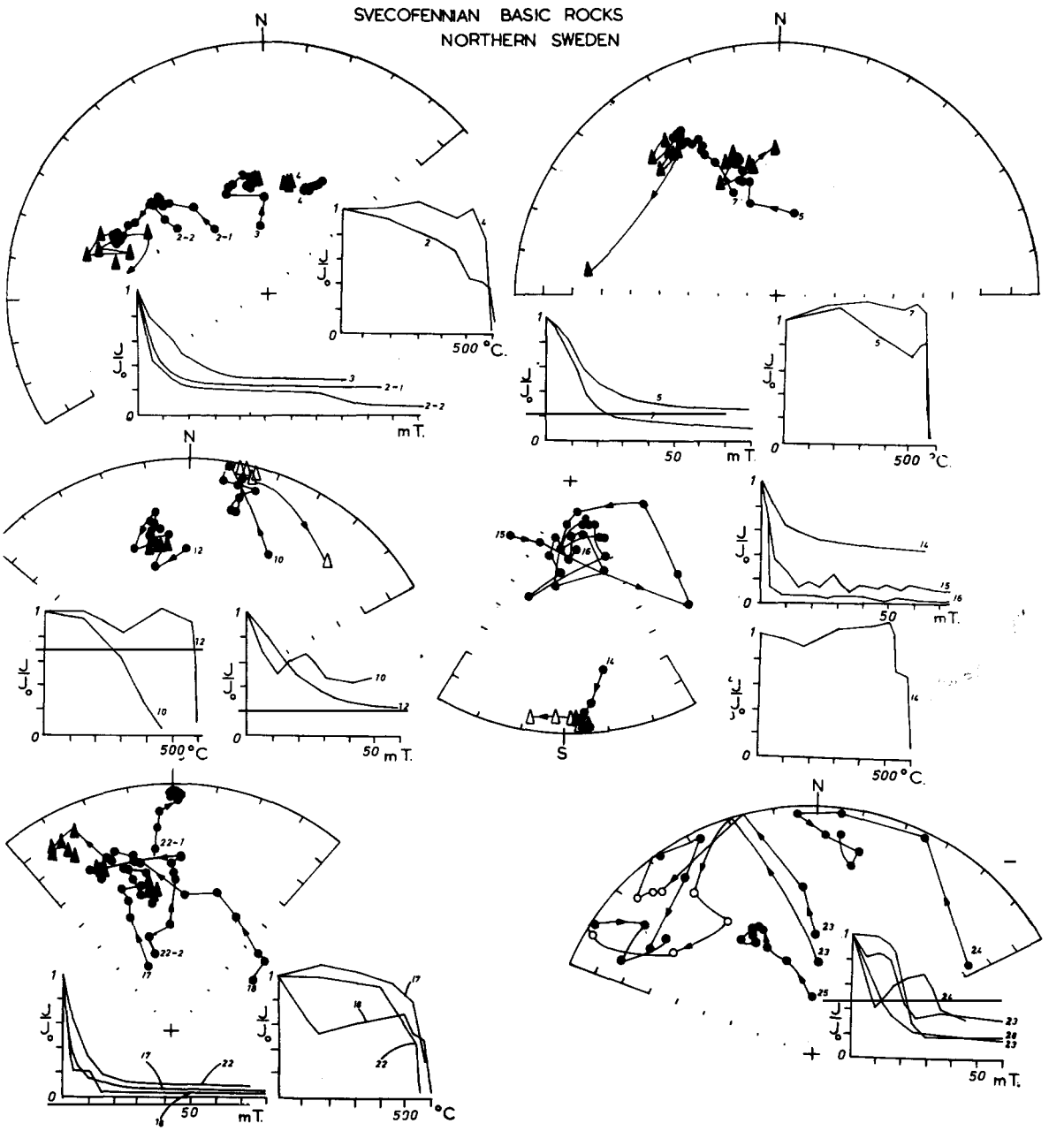


Fig. 2. Behaviour of directions of magnetisation with progressive AF (circles) and thermal (triangles) demagnetisation. The numbers refer to sampling sites (see Fig. 1) and directions are plotted as closed symbols on the lower hemisphere and open symbols on the upper hemisphere (equal area projection). The decay of magnetic moments with progressive treatment is plotted as a fraction of the initial moment (J_0).

(20.9°E, 65.7°N) possess a univectoral component of remanence highly stable to progressive cleaning treatment (Table I and Fig. 2).

Sites 5 to 8 are from the Niemisel Gabbro, a 32 km^2 complex at 22.1°E, 66.0°N. With the exception of a viscous component (<10 mT) in some

samples, only one component is present at all sites and the end points achieved by AF demagnetisation are also stable to subsequent thermal treatment; this component resides in magnetite (Table I). Sites 7 and 8 in dykes cutting the gabbro have identical magnetisations and probably cooled with the major intrusion.

The ~20 km^2 basic intrusion at Sangis on the Baltic coast is a hybrid body foliated in part and incorporating lenses of granitic material. Palaeomagnetic sites are from cuttings in the E4 road and all have weak remanences. Sites 9, 10 and 12 have single-component magnetisations residing in magnetite (Table I). Site 13 exhibits Curie points attributable to pyrrhotite (320°) and magnetite, while site 11 shows only pyrrhotite, which converts in part to magnetite on cooling. Site 11 also exhibits low coercivity and low blocking temperature remanence ($D = 333°, I = -26°$) removed from the magnetite-held remanence at the other sites (mean $D = 356°, I = 26°$); this was probably acquired at a later stage of the regional cooling.

Sites 14 and 16 come from basic rocks in the vicinity of Gällivare, 14 being from a basic intrusion 2 km west of the town and 15 and 16 from amphibolites 4–5 km to the southwest and approximately 500 m higher in altitude. Highly stable magnetite-held remanence at site 14 resides in one component only (Table I) and is of opposite polarity to the earlier sites of this study. Less-stable low-coercivity remanence from sites 15 and 16 also has opposite polarity but steeper inclination (Fig. 2). At the highest stages of AF treatment, four samples move to a poorly-defined mean $D = 20°, I = 27°$; thermal demagnetisation was unsuccessful at these latter sites.

Sites 17–22 are from a 10 km^2 gabbro body exposed as a peninsula on the northern margin of the lake Stora Lulevatten. All sites have a strong low-coercivity remanence (<25 mT) with steep positive inclination. At higher fields, sites 17, 19, 20 and 21 and some samples of 18 and 22 define single stable end points with shallow inclination (e.g., Fig. 2); these are stable to subsequent thermal treatment. A further migration of 10–30° takes place in some samples of site 18 at higher AF or thermal demagnetisation, to define an end point $D = 331°, I = 33°$. A comparable behaviour is observed at site 22 where a shallow inclination direction similar to the remaining sites from this body moves to a steeper northwesterly inclination. This higher blocking temperature component ($D = 343°, I = 33°$) may reside in a haemo-ilmenite because thermal demagnetisation shows a residual component remaining above the Curie point of magnetite.

Sites 23–27 are from small basic bodies in the vicinity of Jokkmokk; sites 23 and 24 are from 10 km west of the town at 19.6°E, 66.6°N, and sites 26 and 27 are from a 10 km^2 body 8–10 km south of the town at 19.8°E, 66.5°N. Three components are present at site 23: in sample with the highest initial remanent intensities a low coercivity northwesterly negative direction of magnetisation (estimated $D = 307°, I = -22°$) is removed in most samples by cleaning in 5–25 mT fields (but survives to higher fields in a few samples), and the remanence moves to a mean $D = 308°, I = 10°$, which survives thermal cleaning to the Curie point of magnetite. In some weakly magnetised samples only one direction is present ($D = 18°, I = -10°$). Thermomagnetic determinations show that both pyrrhotite and magnetite are present here and it is considered probable that the low blocking temperature remanence directions 23(i) and (iii) in Table I both reside in pyrrhotite. Site 24 defines a single weak component comparable to 23 (iii) with low blocking temperature and apparently residing only in pyrrhotite. Sites 25 and 26 are derived from localities some 250 m higher than 24 and 25; number 25 defines a low coercivity remanence (estimated $D = 4°, I = -24°$) present in weakly magnetised samples and a high coercivity remanence ($D = 350°, I = 34°$) in samples with remanent intensities an order of magnitude higher; this survives to the Curie point of magnetite. A comparable sequence of remanence directions is present at site 26; thermomagnetic determinations here show that the low blocking temperature component resides in pyrrhotite.

Site 27 is from the gabbro intrusion at Vuollerim (20.4°E, 66.4°N). Three weakly-magnetised samples retain a direction $D = 359°, I = 42°$, until moments become too small to measure; the remainder start at, or move through, this direction but reach a stable end point at $D = 338°, I = 4°$, which is also stable to subsequent thermal treatment. Site 28 in amphibolites at Harads (21°E, 66.1°N) has weak remanence yielding a direction close to the present field (estimated $D = 3°, I = 75°, k = 11$) with AF cleaning in low fields; with thermal cleaning this moves to a shallower incli-

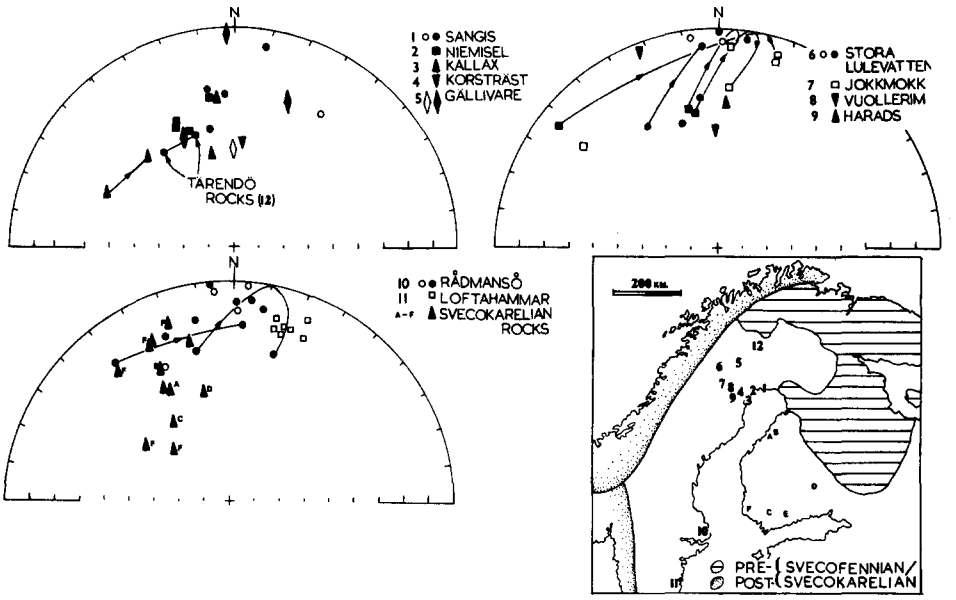


Fig. 3. Directions of magnetisation from basic bodies of the Svecofennian terrain of northern Sweden after cleaning treatment. Open symbols are upper hemisphere plots and closed symbols are lower hemisphere plots, and the arrows indicate the migration from higher to lower blocking temperature components. The inset diagram shows the regional location of sampling localities within the Svecofennian/Svecokarelian terrain; directions of magnetisation from Svecokarelian gabbro–diorites of Finland (Neuvonen, 1974) are also plotted.

nation with poorly-defined mean $D = 3°, I = 31°$.

In summary, the plutonic terrain of northern Sweden exhibits a consistent sequence of magnetisations residing in high blocking temperature constituents (titanomagnetite or haemo-ilmenite) and low blocking temperature constituents (at least in part pyrrhotite). It is characteristic of slowly-cooled plutonic terrains of this kind that directions of magnetisation do not have a truly Fisherian distribution because small differences in critical blocking temperature ranges may result in large differences in the age of magnetisation. Thus sites from the Kallax gabbro show very high individual precisions but cannot be collectively combined, and even some single sites in the Stora Lulevatten gabbro exhibit a narrow range of declinations but a wider range of inclinations. This might result either from (1) incomplete cleaning, (2) acquisition of remanence by a narrow range of blocking temperature spectra over a long period of time during slow uplift, or (3) a broader range of blocking temperature spectra during more rapid uplift. The last two possibilities are favoured for two reasons. Firstly, with a few exceptions, little systematic movement of the remanence vector is observed with thermal treatment close to the Curie point of magnetite. Secondly, when we view the metamorphic province as a whole, the distribution of site mean directions is highly consistent: all of these bodies (distributed over an area 120 X 200 km) define a single sequence of directions (Fig. 3). The sequence of migration from higher to lower blocking temperature directions is always away from northwesterly intermediate positive to northerly shallow positive and shallow negative directions (sites 2, 15, 18, 22, 23, 25, 26, 27); the low blocking temperature components are less well defined but the last part of this sequence is probably represented by low blocking temperature components at sites 25 and 26, and reversed direction at sites 15 and 16.

The highest blocking temperature component cannot invariably be interpreted to be the oldest magnetisation because haematite (unaffected by AF demagnetisation at these ranges and surviving thermal treatment up to 670°C) may in principle result from

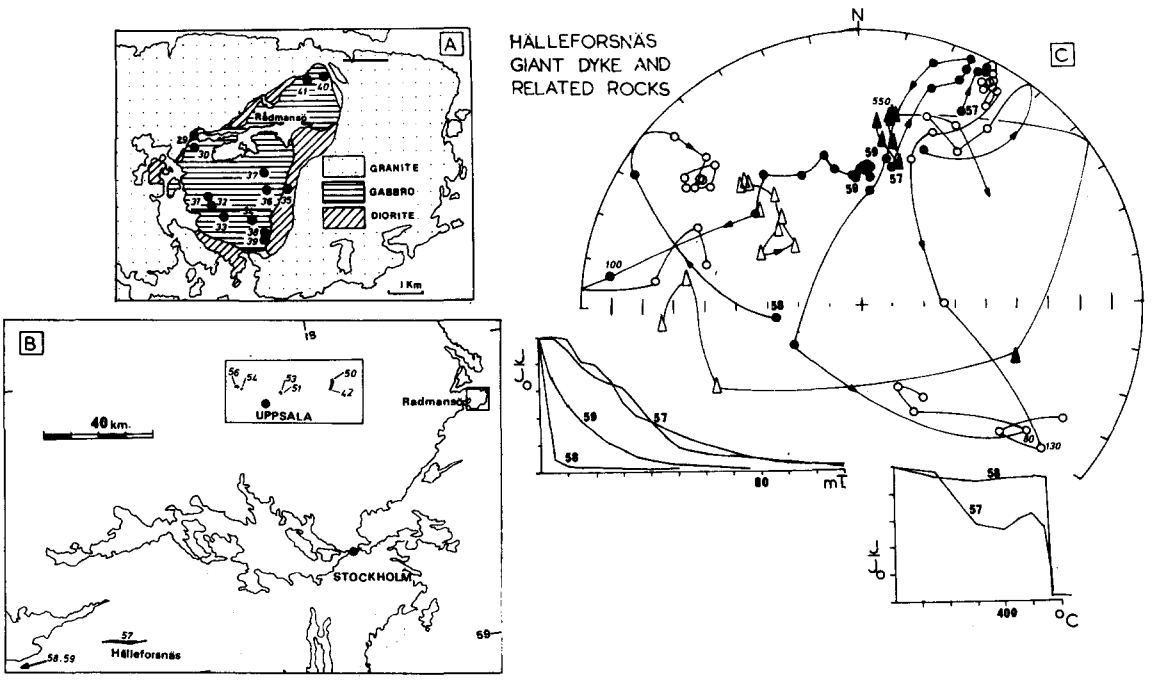


Fig. 4. A: Geology and palaeomagnetic sampling sites in the Rådmansö gabbro–diorite intrusion. B: Regional location of sampling sites in central Sweden; C: Behaviour of samples from the Hälleforsnäs giant dyke and related rocks with progressive cleaning treatment; figures are in mT or °C.

chemical changes during initial cooling or during hydrothermal stages (Buchan and Dunlop, 1976). However, in this instance no single remanence attributable to later remagnetisation is evident and remanence which may possibly reside in haemo-ilmenite is identical to magnetite- or pyrrhotite-held remanence at other sites. Thus the trend of direction changes in Fig. 3 is believed to record the apparent polar movement during the interval of uplift and cooling of this metamorphic province, and interpretation is pursued in the next section.

Also plotted in Fig. 3 are the directions of magnetisation from the Tärendö Gabbro and acidic rocks in the Svecofennian terrain 90 km east of Gällivare (Cornwell, 1968); the gabbro here is intruded by late tectonic granitic rocks dated at 1845–1775 My and has a magnetisation of comparable age to the rocks of this study. The directions of magnetisation from Tärendö fall near the beginning of the time sequence defined from this study. Syn- to post-tectonic gabbro–diorite intrusions are also common in the adjacent Svecokarelian terrain of Finland and Neuvonen (1974) has summarised present palaeomagnetic data relating to these bodies. The distribution of magnetisation directions from Finland overlaps with the field defined by the bodies from Sweden (Fig. 3).

3.3 The Rådmansö complex, eastern Sweden

The Rådmansö gabbro–diorite complex forms an outcrop approximately 18 km^2 in area near the Baltic coast at 19°E, 59.8°N; geology and sampling sites are illustrated in Fig. 4. With progressive AF cleaning, sites 29 and 30 migrate towards northerly shallow inclinations, and additional thermal treatment produces a marginal change in declination in some samples from site 29 but no further reduction in inclination. There are examples where stable end points show a wider spread in inclination (20–30°) than in declination (5–10°), a phenomenon already noted at a few sites in northern Sweden, and the distribution of sample directions in these cases falls along the central part of the swathe of directions in Fig. 3. Sites

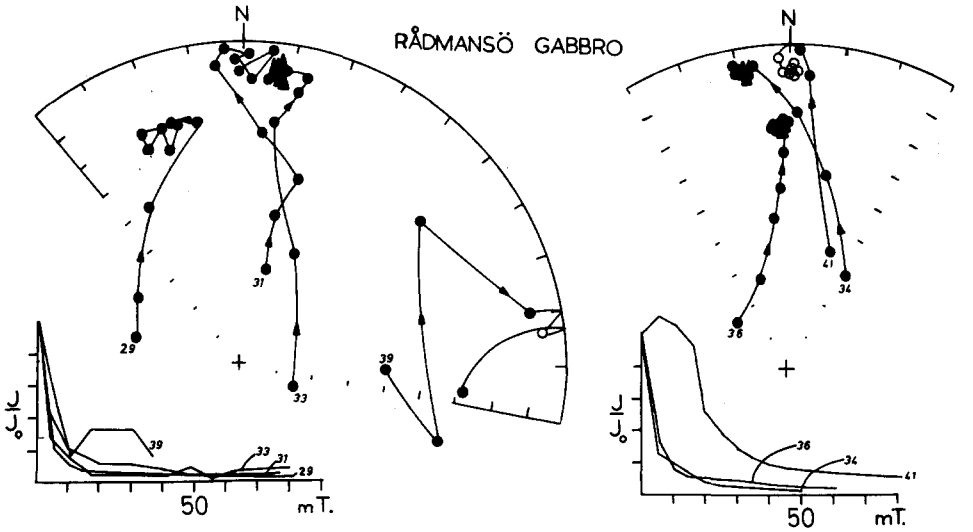


Fig. 5. Behaviour of some samples from the Rådmansö gabbro–diorite complex with progressive AF (circles) and thermal (triangles) demagnetisation; other symbols as for Fig. 2.

31 to 34 migrate to shallow inclinations (Fig. 5) but otherwise have comparable properties to sites 29 and 30, while sites 35–37, again with a wider distribution of magnetic inclinations than declinations, yield mean inclinations falling between these two groups. At site 38 two components are present: 8 samples yield a mean comparable to the remainder of this body while three samples have contrasting declinations (Table II);

TABLE II

Palaeomagnetic results for the Rådmansö gabbro–diorite complex, east Sweden (19°E, 59.7–58.8°N)

Site	*N*	*D*	*I*	*R*	*k*	Palaeomagnetic pole		d*p*/d*m*	Curie point(s) (±10°C)	Curve type	J_0 (× 10^{-5} A m^2 kg^{-1})
						°N	°E				
29(i)	7	347	27	6.90	60	44	216	5/9	555	A	2.44–0.01
(ii)	4	15	27	3.83	17	43	179	14/25			
30	5	339	17	4.93	60	36	225	5/10	565	A	1.95–0.01
31	9	355	–4	8.87	60	28	205	3/7	580	A	2.01–0.25
32	7	4	–1	6.90	61	30	195	4/8	580	A	1.10–0.27
33	5	6	7	4.98	222	34	192	3/5	580	A	1.03–0.55
34	6	354	1	5.96	115	31	207	3/6	580	A	1.32–0.32
35	7	348	15	6.87	46	37	214	5/9	570	A	0.08–0.03
36	8	1	9	7.90	70	35	198	3/7	555	A	13.85–5.09
37	5	9	10	4.82	31	35	188	7/14	565	A	0.43–0.02
38(i)	3	320	16	2.88	17	31	246	17/32	565	A	3.29–0.06
(ii)	8	3	18	7.41	12	39	195	9/17			
39 *	9	106	28	7.50	5	5	88	15/27	560	A	0.13–0.01
40	5	344	–29	4.85	27	13	215	9/17	560	A	0.16–0.01
41	6	2	–11	5.97	148	25	197	3/6	555	A	9.56–1.35
Mean (excluding 38(i), 39–41)											
	11	358.3	11.5	10.7	31	36.1	201.1	4.3/8.5			

* Dyke.

TABLE III

Palaeomagnetic results for the Uppsala metabasite suite, east-central Sweden

Site	N	D	I	R	k	Curie point (°C)	J_0
51	6	0	37	5.96	135	580	0.278–0.053
52	6	352	29	5.93	76	570	0.405–0.063
53	7	5	24	6.96	193	570, 630	0.263–0.079
54 *	8	215	0	7.72	26	575, 630	0.313–0.017
55 *	8	227	9	7.75	28	575	0.579–0.123
56	4	10	31	3.61	8	575, 670	0.332–0.209
						Palaeomagnetic pole	
Group Mean 'A' magnetisation							
	4	1.7	30.6	3.97	90	195°E, 47°N (dp = 6, dm = 11)	
Svecofennian magnetisation, Almunge district							
	16	7.8	8.2	14.3	9	188°E, 34°N (dp = 7, dm = 13)	
Group mean of low blocking temperature ('B') remanence at 9 sites from Almunge district plus site 39 from Rådmansö							
	10	104	9	9.64	25	94°E, –3°N (dp = 5, dm = 10)	

* Sites excluded from group mean.

at sites 40 and 41 sample directions continue to migrate to shallow negative directions with progressive AF cleaning, and thermal demagnetisation again confirms that the AF treatment has achieved stable end points. All of the components listed in Table II are apparently high blocking temperature components residing in titanomagnetite close to pure magnetite in composition. Site 39 is from a later dyke with low coercive force remanence of low stability; magnetic properties are comparable to the dykes with similar trend exposed at Almunge, 50 km to the west (see next section).

The site mean directions of the Rådmansö gabbro–diorite (Table II) fall in a similar field to the magnetisation directions from the Svecofennian terrain of northern Sweden (Fig. 3). At Loftahammar, 250 km to the south of Rådmansö, a comparable gabbro studied by Poorter (1976) has similar directions of magnetisation dated between 1660 and 1460 My and falling close to the younger end of the swathe of directions defined by the collective data from the Svecofennian mobile belt of central and northern Sweden (Fig. 3).

3.4. Uppsala metabasite suite, eastern Sweden

The Uppsala metabasite suite was sampled in three different regions (Fig. 4). Palaeomagnetic analyses of the last two regions (sites 51–56) defines a Svecofennian remanence similar to that from elsewhere: AF demagnetisation yields northerly positive directions with slight reductions in inclination during progressive treatment, followed by no further change. With thermal treatment the bulk of this remanence disappears close to the Curie point of magnetite (Fig. 7). Two sites (54, 55) possess a similar remanence of opposite polarity with slightly shallower inclination than the remainder; this is present as a medium CF * component at site 56 but the vector reverses to join the remaining sites at higher AF fields. With thermal demagnetisation above the Curie point of magnetite, some sample directions at site 55 also reverse via intermediate directions to define a northerly direction with shallow inclination (Fig. 7); this reversed component resides in haematite (Table III). Directions of both polarities at sites 54 and 55 are significantly shallower in inclination than the remaining

* Coercive force.

sites and lie near the younger end of the swathe of Svecofennian directions identified from northern Sweden and Finland (Fig. 3). The remaining sites are combined in Table III to define an "A" magnetisation for the Uppsala metabasite suite.

Nine dykes exposed in the first region near Almunge (18°E, 59.9°N), however, possess quite different remanence. Initial moments, up to an order lower, fall below limits of detectability with thermal treatment at 300–400°C. With AF treatment, samples exhibit erratic changes in moment and poor directional stability; the more stable samples (Fig. 7) move to easterly shallow directions with low median destructive fields. The mean from samples exhibiting this behaviour at 10 sites is $D = 104°, I = 9°$ ($\alpha_{95} = 10°$) but stability is not sufficient to confirm whether this represents a univectorial remanence (Fig. 7). Thermomagnetic spectra show an exponential fall in saturation remanence with no clearly-defined Curie point. This low blocking temperature component has not everywhere displaced a typical Svecofennian remanence which is recovered at samples from a few sites (Table III); in general these samples yield a discrete magnetite Curie point.

The reason for the predominance of low blocking temperature components at this last locality is not completely clear. It does, however, lie close to the margin of the Almunge alkaline complex; this latter body was intruded at 1570 My (Kresten et al., 1977) when the regional Svecofennian rocks were probably still at elevated temperatures, and it is possible that this episode was responsible for physico-chemical effects which largely displaced any original remanence by low blocking temperature components. Pending palaeomagnetic information from the Almunge complex, these are tentatively correlated with a late stage in the regional Svecofennian cooling and are grouped as a "B" magnetisation for this dyke suite in Table III.

3.5. Hälleforsnäs giant dyke and related rocks, central Sweden

Samples from several parts (site 57) of this giant dyke yield a component with northerly shallow "A" directions recovered at a range of demagnetising fields between 5 and 100 mT. This component (Fig. 4) is stable over most of the AF range of treatment and

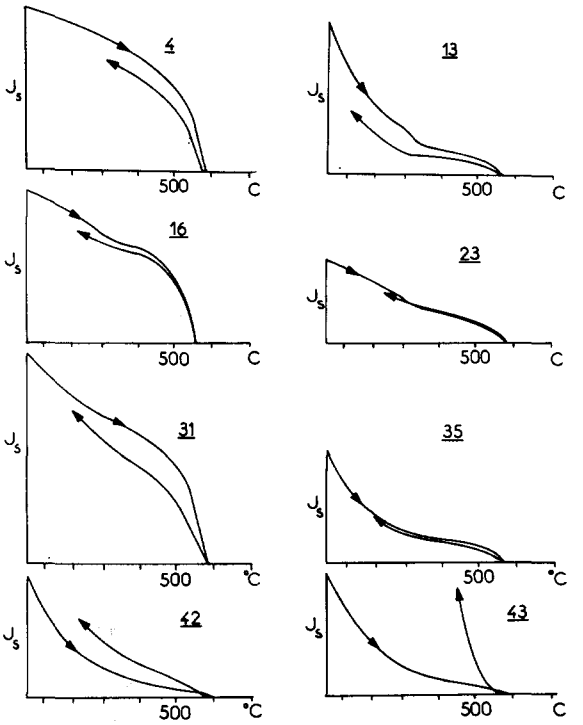


Fig. 6. Thermomagnetic determination (saturation magnetisation, J_s, versus temperature) for some rocks of this study. The J_s axis is arbitrary.

thermal treatment close to the Curie point of magnetite. All sample directions then move to the south-east quadrant with negative inclinations; this direction is only recovered in fields in excess of 130 mT in the highest coercivity samples. The former direction is shown by lower coercivity samples collected near the margin of the dyke, although their directional behaviour tends to become erratic at higher fields. The mean of the higher coercivity ("B") remanence is $D = 116°, I = -33°$ (Table IV). Both haematite and magnetite are observed in thermomagnetic spectra (Table IV) and the highest blocking temperature remanence is of complex origin because at the highest stage of thermal treatment (600–650°C) in some samples inclinations move from negative to positive and declinations reverse. A highest blocking temperature component of approximately $D = 300°, I = -32°$, is indicated here and it appears that the haematite remanence is a composite one with overlapping blocking temperature spectra. The "A"

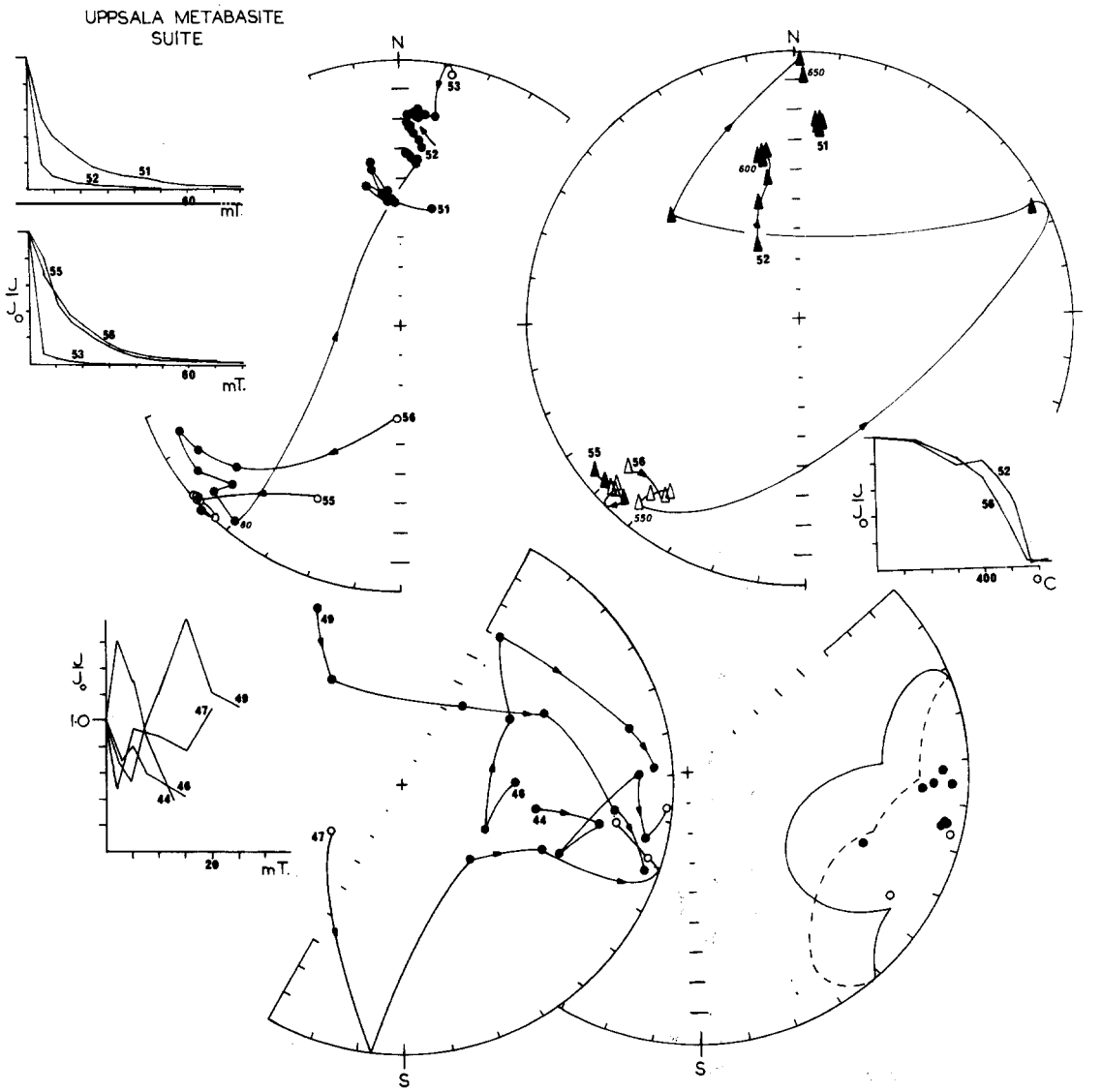


Fig. 7. Behaviour of directions of magnetisation and magnetic moments of samples from the Uppsala metabasite suite exhibiting "A" (upper stereonets) and "B" (lower stereonets) components of magnetisation; symbols as for Figs. 2 and 5.

(magnetite-held) remanence probably approximates most closely to the age of initial cooling, with the secondary components residing in haematite resulting from one or more later episodes of hydrothermal/deuteric activity.

A dolerite body (site 58) nearer the Sveconorwegian mobile zone at Bo (15.5°E, 59°N) has a single magnetite-held component of remanence, $D = 305°$, $I = -41°$ (Fig. 7). This is distinct from the Hällefors-näs dyke but comparable to remanence directions (of both polarities) identified by Patchett and Bylund (1977) from the ~900 My dykes adjacent to the Sveconorwegian front zone, and links the dyke with this system. Svecofennian diorites (site 59) adjoining the dyke outcrop appear to be largely unaffected by this intrusive activity and in the main retain a typical

TABLE IV

Palaeomagnetic results for Hälleforsnäs giant dyke and related rocks

Site		N	D	I	R	k	Curie point(s)	J_0 (A m^2 kg^{-1})
Hälleforsnäs dyke								
57	"A"	13	26.0	−1.6	12.4	21	570, 670 *	56.21–1.25
					(Palaeomagnetic pole 167°E, 27°N; dp = 5, dm = 9)			
	"B"	11	116.1	−32.7	10.3	15		
					(Palaeomagnetic pole 93°E, 29°S; dp = 8, dm = 14)			
Bo dyke								
58		9	305.3	−40.9	8.46	15	580	35.34–0.91
					(Palaeomagnetic pole 244°E, 4°S; dp = 10, dm = 17)			
Svecofennian diorite								
59		8	4.4	46.4	7.87	55	580	5.52–0.32
					(Palaeomagnetic pole 188°E, 59°N; dp = 6, dm = 10)			

* Local, detected in coarse-grained facies.

Svecofennian direction of remanence with AF and thermal cleaning (Fig. 4). In two samples, however, this direction was removed at the highest fields, to recover a direction comparable to the dyke (Fig. 4), illustrating the sporadic acquisition of a high blocking temperature component (presumably residing in haematite) by the later (~900 My) intrusive activity. This northwesterly negative component is close to the highest blocking temperature component in the Hälleforsnäs dyke, consistent with hydrothermal alteration at this later time and the K–Ar evidence of Patchett (1978) for a ~900 My event.

4. Proterozoic apparent polar wander movements relative to the Fennoscandian shield in Lower–Middle Proterozoic times

The present study has greatly extended the range and detail of palaeomagnetic coverage from the Svecofennian mobile terrain, and Table V lists these data together with a compilation of other Middle–Lower Proterozoic palaeomagnetic poles from the Fennoscandian and Ukrainian shields. Although the bulk of this palaeomagnetic record clearly dates from uplift of these widespread ~1800 My mobile terrains, there is some evidence that fragments of the APW record may be preserved from 200–300 My prior to this time. Thus the Jatulian rocks of Finland (pole 9) have a magmatic age of 2150 My although it is possible that the remanence was overprinted by the Svecokarelian activity (Neuvonen, 1975). Also, in a recent revision and summary of palaeomagnetic data from the U.S.S.R., McElhinny et al. (1977) list palaeomagnetic data for the (oldest) Saamo–Karelian zone of the Fennoscandian shield, which come from post-tectonic igneous rocks and sediments assigned to the interval 1950–1600 My (Gerling and Polkanov, 1958; Kratz et al., 1968; poles 1–8 in Table IV). These poles are consistent with the Svecofennian uplift poles (Fig. 8) although their assigned ages are somewhat older. We may conclude either that these estimates are too high, or that the pole was at a broadly similar position during parts of the intervals 1900–1800 and 1700–1600 My; the latter explanation is favoured because these poles agree broadly with the Jatulian rocks of Finland (pole 9 in Table IV) and it would otherwise be necessary to re-evaluate the age assignments of all of them. However, remagnetisation during the Svecofennian/Svecokarelian activity cannot be ruled out.

We have seen that the Svecofennian uplift magnetisations are unlikely to be much older than 1700 My. The youngest magnetite-held remanence is probably the ~1520 My Hälleforsnäs "A", while the Uppsala Metabasite "B" and Hälleforsnäs "B" magnetisations are quite unlike the remaining Svecofennian record (Fig. 8, Table IV) and are presumed to be somewhat younger. Also, rock units yielding magnetisations near the younger end of the swathe of Svecofennian uplift directions in Fig. 3 (and poles in Fig. 8) have comparable magnetisations to certain rocks

TABLE V

A compilation of Lower–Middle Proterozoic palaeomagnetic results from the Fennoscandian shield. All of these units predate the post-Jotnian igneous episode of ~1270–1210 My, for which see Patchett et al. (1978) for a summary of radiometric evidence and Piper (1979b) for summary of palaeomagnetic data.

Pole	Rock unit	Age (My)	Palaeomagnetic pole		dp/dm	Source reference
			°E	°N		
(i) Svecokarelian and pre-Svecokarelian, Russian sector						
1	Central Karelia dolerites	–	235	49	5/7	Khramov, listed by McElhinney et al. (1977)
2	Central Karelia sandstones (i)	1870–1610	239	57	15/21	
3	Central Karelia sandstones (ii)	1870–1610	264	62	5/7	
4	Northern Karelia sandstones	1870–1610	217	65	8/10	
5	Kola Peninsula, gabbro–norite intrusions	1880–1820	223	40	9/16	
6	Onega gabbro–dolerite intrusions	1800–1600	230	32	1/2	
7	Shoksha Group, south Karelia	1950–1850	240	54	3/5	
8	Pedasel'sk and Pukhta Groups	1950–1850	231	44	3/5	
9	Jatulian igneous rocks, east Finland	2150	215	60	–	Neuvonen (1975)
(ii) Ukrainian sector						
10	Korosten Complex *	1750	238	27	–	Kruglyakova (1961), Spall (1973)
11	Uman Granite *	1400–1200	240	26	–	
12	Korosten Granite *	1750	232	24	–	
13	Turchinka Gabbro *	–	212	12	–	
14	Tokova Complex *	1600–1400	326	51	–	
15	Bokovyansk-Osnits *	1700–1500	301	52	–	
16	Ingulets Complex *	2000–1700	325	64	–	
17	Bug Podolsky Complex *	2300–2000	286	54	–	
(iii) Svecokarelian Province, syntectonic basic rocks						
18	Tärendö Gabbro		228	45	10/17	Cornwell (1968)
19	Tärendö acid rocks *		248	42	–	
20	Ylivieska Gabbro		242	43	5/8	Neuvonen (1974)
21	Pohjanmaa gabbro–dolerite		239	38	8/5	
22	Tammela gabbro–diorite		260	52	3/5	
23	Mikkeli gabbro–diorite		230	53	10/15	
24	Hyvinkää gabbro–diorite		246	44	5/9	
25	SW Finland gabbro–diorites **		243	40	10/18	
(iv) Svecofennian province, syntectonic basic rocks						
26	Kallax Gabbro (i)		211	40	2/2	This study
27	Kallax Gabbro (ii)		257	39	4/6	
28	Kallax Gabbro (iii)		279	27	6/10	
29	Kallax Gabbro (iv)		222	57	3/5	
30	Korstrask Gabbro		194	53	6/9	
31	Niemisel Gabbro		227	41	6/10	
32	Sangis Gabbro (i)		207	38	11/20	
33	Sangis Gabbro (ii)		230	8	10/19	
34	Gallivare basic rocks (i)		23	–22	2/5	
35	Gallivare basic rocks (ii)		43	5	8/14	
36	Gallivare basic rocks (iii)		177	34	20/37	
37	Gallivare basic rocks (iv)		20	10	11/16	
38	Stora Lulevatten Gabbro (i)		200	23	4/8	
39	Stora Lulevatten Gabbro (ii)		221	40	11/19	
40	Jokkmokk basic rocks (i)		256	19	6/12	
41	Jokkmokk basic rocks (ii)		189	17	7/13	
42	Jokkmokk basic rocks (iii)		214	41	3/5	
43	Vuollerim Gabbro (i)		202	48	15/25	

(continued)

TABLE V (continued)

Pole	Rock unit	Age (My)	Palaeomagnetic pole		dp/dm	Source reference
			°E	°N		
44	Vuollerim Gabbro (ii)		224	24	5/10	This study
45	Harads amphibolites		197	41	17/30	
46	Rådmansö gabbro–diorite		201	36	4/9	
47	Loftahammar Gabbro	1660–1470	179	23	2/5	Poorter (1976)
(v) Rock units post-tectonic to Svecofennian and Svecokarelian mobile zones						
48	Uppsala metabasite "A"	>1570	195	47	6/11	This study
49	Hälleforsnäs Giant Dyke "A"	1518	167	27	5/9	
50	Uppsala metabasite "B"		94	−3	5/10	
51	Hälleforsnäs Giant Dyke "B"		93	−29	8/14	
53	Häme dolerites, Finland	1800–1640	72	67	14/13	Neuvonen (1967)
54	Kumlinge dolerites, Finland	>1640	204	8	18/28	Neuvonen and Grundström (1969)
55	Föglö Dolerite, Finland	⩾1640	187	31	6/12	
56	Ava intrusives, Finland	1700–1670	169	41	9/17	Neuvonen (1970)
57	Upper-Dala Volcanics, Sweden	<1634	176	23	−/−	Mulder (1971)
58	Jotnian (Dalarna) sediments, Sweden	<unit 53	177	33	7/14	Piper and Smith (1980)
59	Jotnian (Dalarna) lavas, Sweden	<unit 54	189	27	5/11	
60	Jotnian (Satakunta) sediments, Finland	1470–1420	180	3	8/12	Neuvonen (1973)
61	Gävle Granite, Sweden	>1250	165	35	7/16	Piper (1980)
62	Nordingrå gabbro–anorthosite, Sweden	>1385	138	37	4/6	
63	Nordingrå Granite, Sweden	1385	149	33	8/14	
64	Ragunda Complex, Sweden	1293	165	54	6/9	

* Total NRM data.
** Mean of data from 7 localities (Fig. 3).

wich are certainly post-tectonic to the Svecofennian/Svecokarelian mobile belt, and were probably magnetised in high-level crustal environments. These units include the ~1600 My upper Dala volcanics and possibly the overlying Jotnian (Dalarna) lavas and sediments of central Sweden (Piper and Smith, 1980; and Table IV) although these can only currently be assigned to the broad interval 1600–1300 My. The Kumlinge dolerites and Foglö dolerite of Finland which are associated with, or older than, the type-Rapakivi magmatism (1640 My, Gerling and Polkanov, 1958; Neuvonen and Grundström, 1969). This agreement suggests that the bulk of the Svecofennian/Svecokarelian rocks acquired their magnetisations at a late stage in the uplift of these terrains. This is likely both because blocking temperature ranges are lowered and broadened by protracted cooling (Pullaiah et al., 1975), and because the uplift rates of pre-1000 My mobile zones (Watson, 1976) were slow, and magnetisation temperatures would have been considerably lowered (e.g., Morgan, 1976; McWilliams and Dunlop, 1978).

The distribution of poles running from east to west in Fig. 8 is defined by the collective distribution of palaeomagnetic directions (Fig. 3) running from northwesterly or northerly positive to northerly shallow or negative, and the consistent migration from high to low blocking temperature directions defines this as a migration direction of the palaeofield. This is further indicated by the systematic motion between high and low blocking temperature components in individual rock units which also define part of a loop at the younger end of the east–west swathe in Fig. 8; high blocking temperature components represented by poles 40 and 28 may define the earliest part of the motion (Fig. 8) but it is not established that they are older than the other high blocking temperature components. Age data relating to the poles of Table V also permit some refinement of the younger part of this path: Finnish poles 54–56 have assigned ages of ~1650 My and, together with pole 57 (~1600

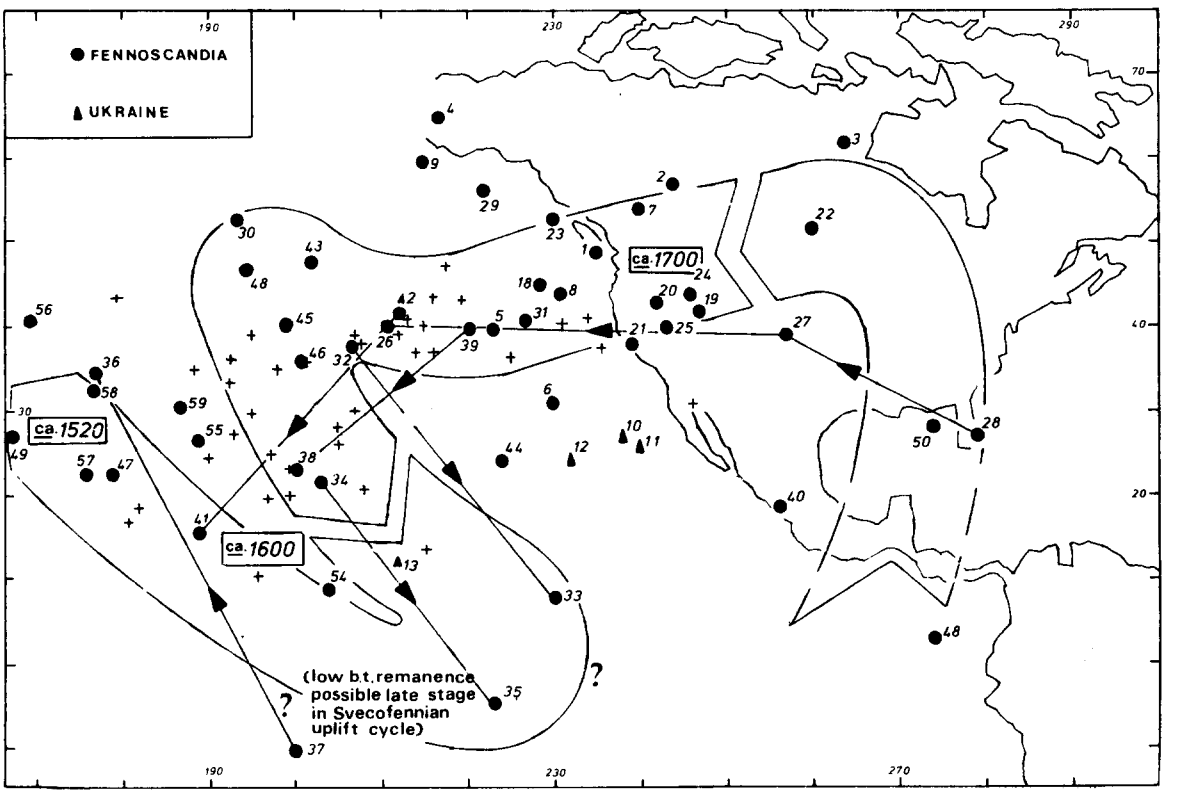


Fig. 8. Middle Proterozoic palaeomagnetic poles from the Fennoscandian shield (~1700–1550 My). The solid swathe indicates a generalised apparent polar wander path for this interval as a band 15° wide derived from the distribution of remanence directions illustrated in Fig. 3. This may not completely embrace polar movements during this interval (see text). Crosses are site means (Table I).

My), suggest that a closed loop is present in the APW path prior to pole 49 (1520 My). Indeed, the association of the Finnish poles with poles from northern Sweden associated with the younger Lina Granite suggests that the path was at a similar position at ~1650 and ~1520 My. The alternative possibility is that the Finnish rocks have late-Svecofennian magnetisations post-dating the (mostly Rb–Sr and U–Pb) age assignments. More data will be necessary to resolve this point and may require a refinement of the path indicated in Fig. 8.

It is next pertinent to enquire whether the magnetisations in the Svecofennian terrain (Figs. 3 and 8) give a complete record of the apparent polar movements between ~1700 and 1500 My. It seems most likely that this record is incomplete for three reasons. Firstly, magnetisation records from basement terrains tend to be incomplete both because uplift may have been episodic, and because the blocking temperature ranges of the magnetic minerals (although broadened by protracted uplift) are relatively narrow in comparison with the total temperature ranges passed through during uplift. Secondly, sites 15, 16 and 23 yield high blocking temperature components which are not readily interpreted in the context of the remaining Svecofennian record. Thirdly, the Häme Dolerites of Finland (Neuvonen, 1967), with a probable age between 1800 and 1640 My, also yield a palaeomagnetic pole (49 in Table IV) quite unlike other poles assigned to this interval [Although the magnetisation direction is close to the present field, the cleaning employed by Neuvonen was almost certainly sufficient to eliminate this as a possible explanation. (See also Lapointe et al., 1978)]. Thus, although the present data listed in Table IV show that APW movement relative to the

Fennoscandian shield between ~1700 and 1530 My can be generalised by a large open loop (Fig. 8), in the shorter term APW movements were probably more complex. This is apparent both from the distribution of poles from individual sites (these can be regarded as palaeomagnetic poles because their extended blocking temperature ranges were certainly long enough to average secular variation), and from those poles which do not fit this generalised path.

Kruglyakova (1961) reported several poles from Proterozoic rocks of the Ukrainian shield. It seems likely that these were calculated from total NRM data (Spall, 1973) which may include later components and they cannot be regarded as definitive data comparable with the other poles of Table V. The review of Ukrainian geochronology by Semenenko et al. (1968) suggests that poles 10–13 (Table V) represent an age of 1750 My and they plot within the field of the Svecofennian poles. Poles 14–17 are more difficult to interpret (Spall, 1973) and, on the basis of their age assignments, may represent any time between 2300 and 1400 My; these do not correspond with the Svecofennian/Svecokarelian poles.

5. Mid-Proterozoic configuration of the Fennoscandian and Laurentian shields

General correlations between the fields of mid-Proterozoic (1700–1200 My) palaeomagnetic poles from the Fennoscandian and Laurentian shields suggest that they were in juxtaposition during this interval and formed a single coherent continental unit (Piper, 1976; Poorter, 1976). This correlation is best defined over the interval 1450–1200 My during which the APW records of the two shields define a closed loop (Piper, 1980 and Fig. 10). By superimposing these two loops we define an optimum reconstruction with a resultant rotation of approximately 64° about a Euler pole at 1°E, 36°N. The pre-1450 My Fennoscandian palaeomagnetic data of Table V are compared, after rotation about this Euler pole, with the contemporaneous Laurentian data in Fig. 9A. This operation is seen to achieve a close correspondence between the Svecofennian/Svecokarelian poles and palaeomagnetic data from the ~1800 My Hudsonian terrain of the Laurentian shield. It emphasises the close relationship in both space and time of the large areas of crustal mobilisation at ~1800 My (Fig. 9B) with uplift and cooling extending over periods of 100–300 My after this (Magnusson, 1960; Stockwell et al., 1970; Welin et al., 1971). There is a notable correspondence between the loop defined by the sequence of magnetisations in the Coronation geosyncline (McGlynn and Irving, 1978) and uplift magnetisations in the Churchill terrain and West Greenland (see summary in Cavanaugh and Seyfert, 1977 and Piper, 1979a), and the sequence of magnetisations defined from the Svecofennian/Svecokarelian terrain (Fig. 9A).

A smaller body of data from the (1800–1600 My) Ketilidian, Naggsugtoqidian and Lewisian terrains of Greenland and northwest Scotland (Beckmann and Mitchell, 1976, Piper and Stearn, 1976; Piper, 1979a) also define an extension to the Laurentian loop at ~1650–1600 My but this may represent a relatively short excursion because it has not yet been found more widely represented in the Laurentian shield. The single Fennoscandian pole 49 (Fig. 9A) suggests that the same excursion is recorded in that shield, but since the age of this pole is rather poorly defined (Neuvonen, 1967), the APW path of Fig. 9A has not been extended to include it.

The younger part of the Fennoscandian APW path of Figs. 8 and 9A is not known with certainty to correlate with any Laurentian data, but the inadequacy of the 1600–1450 My record from the Laurentian shield does not permit a meaningful test here. The younger (~1600–1500 My) magnetisations from the Lewisian terrain of northwest Scotland define a field which is broadly comparable with this younger segment of the Fennoscandian path (Fig. 9C) but uncertainties surround the precise configuration of the Lewisian region within the Laurentian shield (Piper, 1979a) and do not, at present, permit a precise correlation.

6. Comparison with the Siberian shield and general conclusions

The Urals are widely recognised as an orogenic belt defining the confluence and welding of the Fennoscandian and Siberian shields in Phanerozoic times (Hamilton, 1970; Windley, 1977), and there are no tectonic grounds for attempting a direct correlation

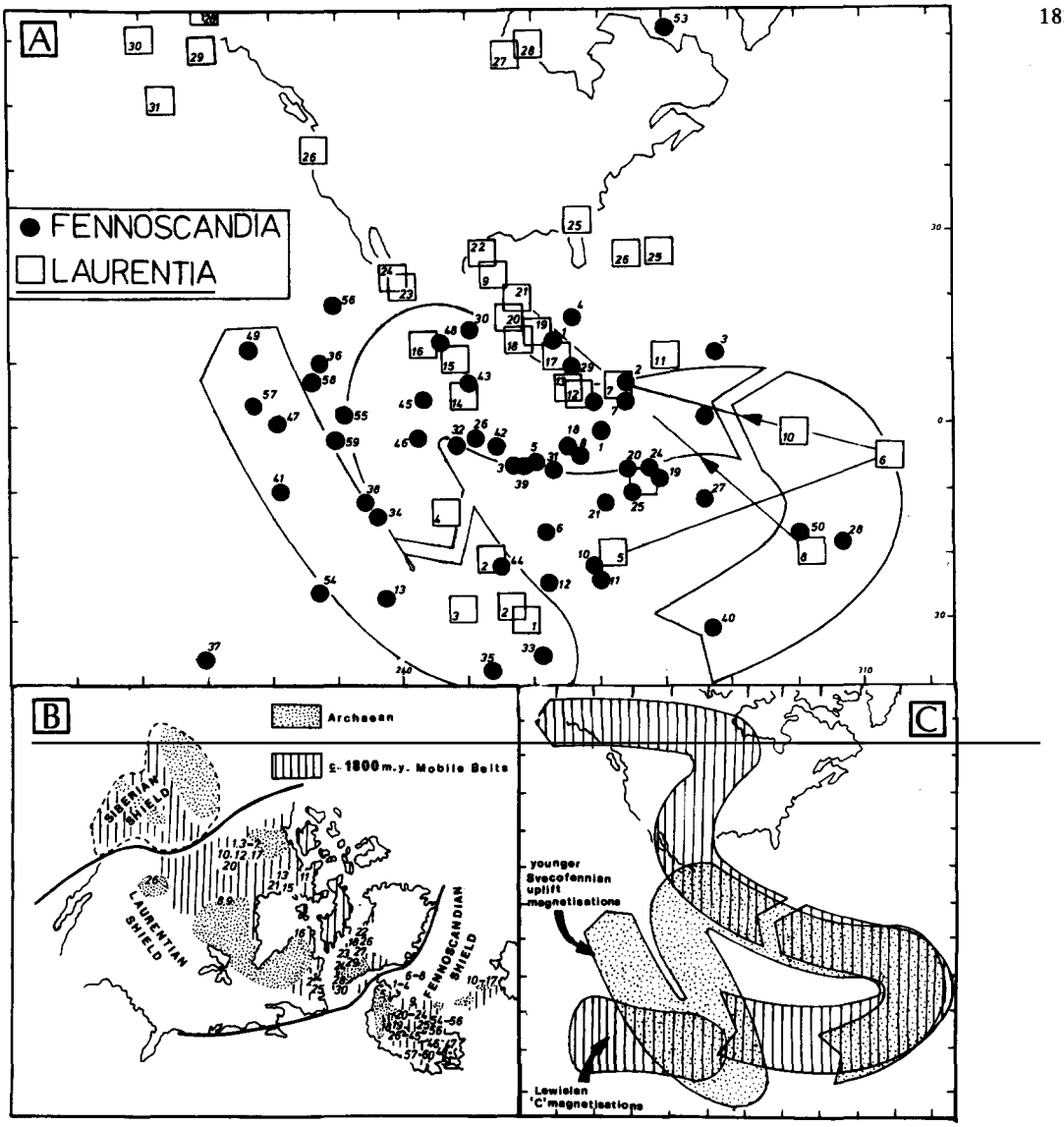


Fig. 9. A: The Middle Proterozoic palaeomagnetic poles (circles) from the Fennoscandian shield (Table IV and Fig. 8) rotated and compared with the contemporaneous poles from the Laurentian shield (squares) according to the reconstruction derived from 1200–1450 My data (Piper, 1976, 1980; Poorter 1976 and Fig. 10). The Fennoscandian APW swathe is the same as that indicated in Fig. 8 and the Laurentian data and APW curve is based on the compilations of McGlynn and Irving (1978), Cavanaugh and Seyfert (1977) and Piper (1979a). The Laurentian poles (with ages in parentheses) are numbered: 1, Peninsula Sill (1800 My); 2, Seward Subgroup (1800 My); 3, Douglas Peninsula Formation, "A" magnetisation; 4, Takiyuak Formation; 5, 6, 7, Pearson Formation, "A", "B" and "C" magnetisations; 8, 9, Flin Flon secondary (1800) and tertiary (1700–1600 My) magnetisations; 10, Et-Then Group (1835–1630 My); 11, Sparrow dykes (1700 My); 12, Merihek "B" magnetisations; 13, Dubawnt Group (1787 My); 14, Seward "B" magnetisations; 15, Daly Bay "A" magnetisations; 16, Cape Smith metavolcanics (1650–1460 My); 17, Nonacho sediments (>1700 My); 18, 19, Itivdleq area dykes (1830–1712 My); 20, Kahocella secondary (<1873 My); 21, metamorphosed Kaminak dykes (1892–1615 My); 22, Nordre Stromfjord basement; 23, Kangamiut dykes (1790–1620 My); 24, Godhåb gneisses; 25, Retty "B" magnetisations; 26, Holsteinsborg basement; 26, Wind River dykes (1880–1680 My); 27, Sagdlerssuaq basement (1620 My); 28, Ketilidian metavolcanics (1730–1600 My); 29, West Greenland amphibolite bondins; 30, Lewisian "A" magnetisations; 31, Sarfangaqualand amphibolites (1620 My). B: The areal distribution of palaeomagnetic studies yielding the poles of Fig. 9A, plotted on the reconstruction of the shields; the terrains affected by the ~1800 My mobile episodes are stippled. C: The 1750–1550 My apparent polar wander paths for the Fennoscandian (stippled) and Laurentian (vertical shading) shields on the reconstruction of Fig. 9B as defined by the present data (see text).

of the Precambrian palaeomagnetic record of the two shields. However, on purely geological evidence, Sears and Price (1978) observed a close correlation between the Siberian shield and the western sector of the Laurentian shield; their reconstruction yielded a match of the geometric margins and a congruence of both tectonic fabric and age provinces. They did not attempt to determine whether their reconstruction was in accord with the palaeomagnetic evidence, but a new summary of data from the U.S.S.R. (McElhinny et al., 1977, based on compilations of Khramov) does permit a useful test to be made.

Geometrical analysis of the reconstruction of Sears and Price shows that it requires an anticlockwise rotation of approximately 69° about a Euler pole at 326°E, 55°N. If this operation is performed on the Siberian (Precambrian) palaeomagnetic data (Table VI) it achieves the match with the contemporaneous Laurentian and Fennoscandian APW record illustrated in Fig. 10B.

The Siberian shield comprises large areas of terrain defined by linear magnetic anomalies (Garfarov, 1970) which had stabilised by 1750–1700 My following deformation between 2100 and 1800 My (Kosygin and Parfenov, 1975); they separate relict Archaean (Aldan, Tungusska and Tyunga) massifs and are contemporaneous with the comparable Hudsonian and Svecofennian/Svecokarelian activity (Kosygin and Parfenov, 1975; Sears and Price, 1978). Only pole 1 (Table V) from the Olenek gabbro–dolerites appears to represent this interval and corresponds with the Svecofennian poles (Fig. 10A). All the remaining poles (with the exceptions of 2 and 7) are from sedimentary sequences representative of platform sedimentation that commenced at about 1570–1450 My and continued to late Precambrian times. Age estimates for these poles are accordingly only approximate (Fig. 10B and Table V) but the poles do define a general grouping of Upper Proterozoic age (~1300–900 My) which coincides closely with the Laurentian APW record of ~1270–1200, 1000 and 850 My (Fig. 10A). This palaeomagnetic comparison thus provides some positive evidence for accepting the reconstruction of Sears and Price, although on the debit

TABLE VI

Siberian shield, Precambrian palaeomagnetic results

Pole	Rock unit	Age (My)	Palaeomagnetic pole		dp/dm	Pole rotated relative to Laurentian shield	
			°E	°N		°E	°N
1	Olenek region, gabbro–dolerites	>1600	274	32	4/8	306	0
2	Aldan region, intrusions	–	264	3	9/15	327	–17
3	Khapchan Series (i)	–	304	12	8/15	174	–14
4	Khapchan Series (ii)	–	305	16	7/13	172	–18
5	Yenisei Group	<1540	272	49	3/6	307	–25
6	Uluntui Group, northern Baikal	<1570	263	29	4/8	312	5
7	Northern Anabar shield dolerites	1135–820	203	10	3/5	261	–28
8	Karagassk Group, Sayan region	870–840	321	–22	3/6	208	3
9	Omninsk Group	–	203	–30	3/5	257	–68
10	Malginsk Group	–	217	–21	3/4	284	–58
11	Tsipanda Group	–	208	–24	4/8	268	–62
12	Lakhanda Group	890	203	–23	2/4	259	–61
13	Gorbilok Group	1320–1270	278	13	9/17	153	1
14	Uderei Group	1320–1270	282	20	4/7	152	–7
15	Pogoryui Group	1150–1120	286	20	6/10	155	–9
16	Potoskui Group	1100–1050	295	25	6/11	158	–18
17	Shuntar Group	950–930	296	18	9/17	164	–13
18	Kirgitei Group	930	290	17	4/2	160	–9
19	Nizhneangara Group (i)	930–744	322	10	8/15	190	–23
20	Nizhneangara Group (ii)	930–744	269	17	9/14	143	2

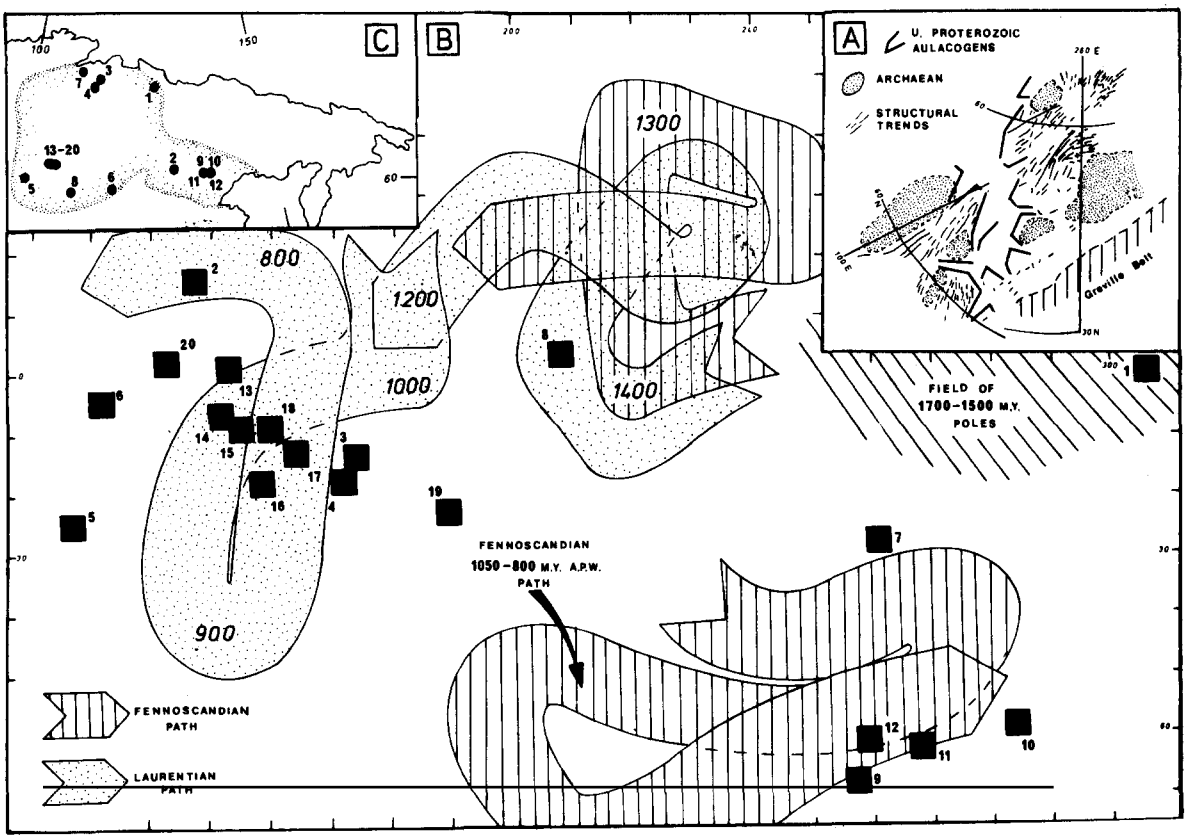


Fig. 10. A: Proterozoic reconstruction of the Laurentian and Siberian shields according to Sears and Price (1978). B: The Proterozoic palaeomagnetic data from the Siberian shield (Table V) compared with the Fennoscandian and Laurentian Upper Proterozoic APW paths according to this reconstruction. The Fennoscandian and Laurentian APW paths are compiled from Irving et al. (1977). McWilliams and Dunlop (1978) and Piper (1980). The APW loop variously called the "Great Logan" of "Keweenawan" loop and rapidly executed between 1180 and 1100 My is excluded from this diagram. The large figures are ages assigned to these APW paths in My and the small figures refer to the Siberian poles of Table V. C: The location of the Siberian palaeomagnetic studies within this shield.

side we note an absence of correlation between Siberian poles 7 and 9–11 and the Laurentian/Fennoscandian data (the latter group may be post-800 My in age and therefore not relevant to the comparison of Fig. 9). Comparison of pre-1200 My and post-1050 My palaeomagnetic poles from the Laurentian and Fennoscandian shields (Patchett and Bylund, 1977; Patchett et al., 1978) define a rotational movement between the two shields such that the configurations of pre-1200 My and post-1050 My APW paths are quite different (Fig. 10A); when the relationship between the earlier and later Proterozoic polar path for the Siberian shield is defined it should be possible to use this to identify with certainty a link with either the Laurentian or the Fennoscandian shield.

General implications of the continental configurations have been discussed elsewhere (e.g., Davies and Windley, 1976; Sears and Price, 1978; Piper, 1980) and include (1) a continuity of terrains mobilised at ~1800 My (Fig. 9C) and 1100 My, (2) an alignment of many tectonic lineaments, and (3) a contraction of anorthosite–rapakivi type magmatism following the ~1800 My mobile episodes towards one margin of the shields. Siberian Proterozoic magmatism may also conform with these observations on the Sears–Price reconstruction. Older sporadic anorthosites in the north-western part of the shield were deformed by the ~1800 My activity and the frequency of massive

anorthosites increases towards the eastern sector (including intrusion at ~1700 My) while alkali granite and syenite magmatism continued here until 1100–1000 My (Tugarinov, 1968; New York State Museum Science Service, 1969). However, more age data are needed to confirm this.

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